

Jesse Tsang

Full-Stack Software Engineer | AI-Assisted Workflow Builder | Angular / TypeScript / Java

Montreal, QC | Canadian citizen, authorized to work in Canada | No sponsorship required

JesseTsang09@gmail.com | 438-995-9671 | linkedin.com/in/jesse-tsang

Professional Summary

Full-Stack Software Engineer with 6+ years of experience building production web applications, backend services, automated test suites, CI/CD workflows, and operational tooling across Angular, TypeScript, Java Spring Boot, Node.js, Python, PostgreSQL, and cloud-connected systems. Strong track record turning ambiguous product and operational requirements into reliable, testable, documented software across transportation, financial services, regulatory reporting, HR modernization, and enterprise tooling.

Currently applying AI-assisted engineering methods to workflow design, documentation, implementation planning, code review, test authoring, and security review. Recent work includes a public AI Ops application-lab project that treats a job application as an auditable workflow: role-source discovery, role intelligence, CV tailoring, skeptical review, ATS form inspection planning, automation boundaries, and human approval gates.

Core Strengths

- Production full-stack engineering with Angular, TypeScript, Java Spring Boot, Node.js, REST APIs, PostgreSQL, and Python
- Operational automation, internal tooling, and workflow design for high-friction business processes
- SQL, relational data modeling, data pipelines, reporting workflows, and test data generation
- CI/CD, quality gates, automated test strategy, release validation, and production readiness
- Production troubleshooting, L3 support, incident triage, and cross-functional stakeholder communication
- Documentation, runbooks, architecture explanation, and practical human-in-the-loop AI workflow design

Technical Skills

Frontend: Angular, TypeScript, RxJS, NgRx, Reactive Forms, Angular HttpClient / Interceptors, HTML5, SCSS/CSS3, AG Grid

Backend: Java 17, Spring Boot, Spring Batch, REST API design, Scala microservices, Node.js / Express

Testing and QA: Jest, Cypress, Playwright, Postman automation, ATDD, regression testing

Data: PostgreSQL, SQL, relational data modeling, Hibernate / JPA, test data automation

Cloud, DevOps, and Observability: AWS RUM, Azure, Docker, GitHub Actions, Jenkins, SonarQube, JFrog Artifactory, XL Deploy, Git

AI-Augmented Engineering: GitHub Copilot, Claude Code, Cursor, MCP integrations, Replit, LLM-assisted architecture planning, implementation support, security review support, automated test authoring, AI workflow documentation

Design and Prototyping: Figma, component-driven UI planning, responsive UI design, practical UI/UX principles

Accessibility: WCAG 2.1 / A11Y, semantic HTML5, keyboard navigation, ARIA

Programming Languages: Python, JavaScript

Spoken Languages: English, French, Cantonese, Mandarin

Current AI Ops Workflow Project

Opendoor AI Ops Application Lab

AI-first application workflow project | May 2026

- Designed an AI-assisted workflow to treat a job application as an operations process: source discovery, role analysis, application strategy, CV tailoring, skeptical review, ATS inspection planning, safe automation boundaries, and retrospective documentation.
- Applied current personal-project experience with Claude Code, Cursor, MCP integrations, and Replit to refine AI-assisted development workflows, prompt strategy, documentation structure, and human-in-the-loop review patterns.
- Used AI to locate and verify the correct Opendoor Operations AI Engineer job description through the employer ATS and corroborating sources, then converted the role into a structured intelligence report and application strategy.
- Established safety controls for truthfulness, privacy, consent, and final submission: private seed CV inputs remain local, generated claims must trace to source facts, and any ATS automation must pause before consent or final submission.
- Structured the repository as a reviewable process artifact with decision logs, execution tracking, prompts, inputs, outputs, evidence folders, and working guidelines for future AI sessions.

Professional Experience

CGI Inc. - ViaRail New Reservation System

Full-Stack Engineer, Angular / TypeScript | March 2026 - Present | Montreal, QC, Remote

High-traffic national passenger booking SPA supporting B2C and B2E booking, payment, fare selection, group search, and booking modification journeys across Canada and the U.S.

- Architected and shipped a production-safe Angular `HttpInterceptor` to prevent double-payment race conditions on `POST /payment`, using RxJS request coalescing, TTL-based response caching, and declarative endpoint rules.
- Released the interceptor behind a feature flag with AWS RUM telemetry and presented architecture and security review details to client architects.
- Validated the interceptor with 37 ATDD-driven automated tests across 5 suites covering request coalescing, TTL expiry, LRU eviction, interceptor-chain integration, and configuration acceptance criteria.
- Resolved a B2C modify-flow payment orchestration regression where non-main-contact loyalty members could access restricted payment methods by consolidating eligibility logic, fixing RxJS startup behavior, correcting cache keying, and adding timeout/retry resilience.
- Identified a frontend-only security enforcement gap in payment restrictions and coordinated server-side enforcement with the backend team.
- Diagnosed and fixed a group-search sleeper fare matrix failure by correcting split-API trigger logic from aggregate inventory to per-cabin configuration, validated with boundary-aware unit tests.
- Fixed a multi-fare upgrade refund integrity bug where merge logic dropped replacement item IDs and caused incomplete refunds or stale Review & Pay summary line items.
- Delivered Angular UI features in compliance with ViaRail's WCAG 2.1 A/AA accessibility mandate using semantic HTML, ARIA, keyboard navigation, and screen-reader-compatible patterns.
- Coordinated REST API contract updates with backend services to align frontend feature requirements.

Key technologies: Angular 20, TypeScript 5.8, RxJS 7, Jest, AWS RUM, REST APIs, WCAG 2.1

CGI Inc. - Desjardins Enterprise Loan System Modernization

Software Developer | June 2025 - December 2025 | Montreal, QC

Greenfield enterprise lending platform used by Desjardins loan officers across multiple business units.

- Owned full-stack feature delivery from technical design through QA, building Angular reactive UI modules backed by Java Spring Boot and Spring Batch processing pipelines.
- Built AG Grid data tables, Reactive Forms with complex validation, REST-backed flows, and backend batch logic with responsibility for correctness, performance, and production readiness.
- Collaborated with business analysts and product managers to refine requirements, propose technical approaches, and break down features into testable increments.
- Used Figma during feature delivery and stakeholder discussions to understand UI expectations, align implementation details, and translate design intent into Angular components.
- Spearheaded QA automation with Postman scripted test collections and Cypress end-to-end suites that became the shared release validation baseline.
- Automated multi-schema test data generation with Python-generated SQL inserts, reducing QA environment setup time and enabling parallel developer testing.
- Mentored junior engineers through regular code reviews, improving team quality standards and ramp-up on Angular and Spring patterns.

Key technologies: Angular 15, TypeScript, RxJS, Java Spring Boot, Spring Batch, PostgreSQL, Cypress, Postman, Docker, SonarQube, Git, Figma

CGI Inc. - HR Application Modernization

Software Developer | April 2024 - June 2025 | Montreal, QC

Modernization of a legacy IBM/i HR system into an Angular and Spring Boot architecture serving internal HR teams.

- Delivered Angular SPA modules end to end, including routing, reactive forms, RxJS-driven component state, and integration across new Spring Boot APIs and legacy Thymeleaf templates.
- Maintained Cypress test suites as the primary functional parity guarantee during a high-risk live migration from legacy to modernized modules.
- Enforced quality gates with SonarQube, coordinated artifact promotion with JFrog, and managed deployment pipelines with XL Deploy.
- Owned the CI/CD loop for team deliverables to support repeatable release validation and production readiness.

Key technologies: Angular 15, TypeScript, RxJS, Java Spring Boot, Cypress, IBM/i, SonarQube, XL Deploy, JFrog

CGI Inc. - Airbus A220 Path Management

Lead Software Developer | January 2024 - March 2024 | Montreal, QC

Short-mandate technical lead role focused on automated migration tooling and stakeholder validation.

- Proposed and implemented a Python-based automated document path migration tool for Bombardier-to-Airbus migration workflows, owning the full lifecycle from solution design to on-site validation.
- Built a dual-strategy processing pipeline using dictionary normalization and regex with CLI tooling and modular libraries designed for reuse.

- Achieved approximately 10,000 files/hour throughput, materially reducing manual migration effort.
- Authored user-facing documentation and conducted on-site validation with non-technical stakeholders to support adoption.

Key technologies: Python, CLI tooling, unit testing, technical documentation

Societe Generale - CAT/CAIS Regulatory Reporting

Java Backend Developer | October 2021 - December 2023 | Montreal, QC

Microservices-based regulatory reporting platform for U.S. markets processing trade data at scale under strict compliance and uptime requirements.

- Designed and maintained Java Spring Batch and Scala microservices for high-volume regulatory reporting pipelines.
- Consumed and produced REST APIs integrated with compliance monitoring systems, owning work from design through production support.
- Provided L3 production support for New York operations, triaging incidents, clarifying requirements with business stakeholders, and delivering fixes under time pressure across time zones.
- Managed Jenkins CI/CD pipelines, code review workflows, SonarQube quality gates, and JFrog artifact management to support reliable releases.
- Ran Agile ceremonies as interim Scrum Master, including sprint planning, grooming, retrospectives, and stakeholder demos.
- Conducted structured code reviews and knowledge-sharing sessions to raise team standards across the pipeline codebase.

Key technologies: Java Spring Boot, Spring Batch, Scala, Python, PostgreSQL, Azure, Jenkins, Docker, SonarQube, JFrog, Angular

Nuance Communications - Enterprise Access Management

R&D Developer Intern | September 2019 - April 2020

- Built internal access management tooling with Spring Boot, HTML/JavaScript front-end, SSO/Keycloak integration, and role-based access control.

Education

Bachelor of Computer Science, Concordia University, Montreal | 2020

Specialization: Web Services and Applications

Notes

CGI client mandates are delivered through CGI Inc. This CV intentionally frames AI experience as current AI-assisted workflow practice and project work, not as long-running historical AI production experience. It does not claim LLM deployment or model drift monitoring experience.